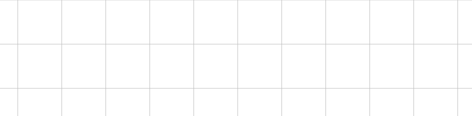
S^n, ~~$S^1\times S^1$~~, $\mathbb{RP}^n$,



$(n\geq 1$
$\downarrow$
$\pi_1=0$.

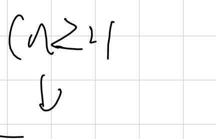
2-dim mfd

$L(p,q)$

$M \# N$. $\wedge$ $\vee$

~~$(D^1\times S^1)/e^{2\pi i p}, e^{2\pi i q}$~~

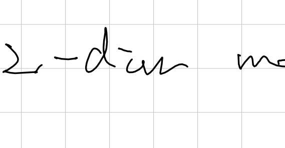
$[0,1]^2$



torus.

$\mathbb{Z}\times\mathbb{Z}$

Klein.



Klein. free product.

~~$\mathbb{Z}\times\mathbb{Z}_2$~~

ababab b..

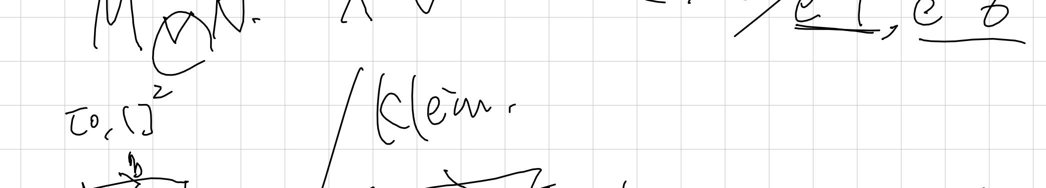
$\langle a,b \mid aba=b\rangle$.

$aba^{-1}b^{-1}=e$.

$\langle a,b \mid ab=ba\rangle$
$\mathbb{Z}\times\mathbb{Z}$

$S^1\vee S^1$

$ab=ba \rightarrow$ group + I.



$S^1\vee S^2$

T: $\mathbb{Z}\times\mathbb{Z}$

$S^1\vee S^1$
$\mathbb{Z}*\mathbb{Z}$

T^1: $\mathbb{Z}*\mathbb{Z}$

K^1: $\mathbb{Z}*\mathbb{Z}$

Theorem

$S^1\vee S^1$
$\mathbb{Z}*\mathbb{Z}$

S^1
$\updownarrow$
$\mathbb{Z}$

theorem

S^3: 0.
$\downarrow$
$\mathbb{R}^3$: 0.

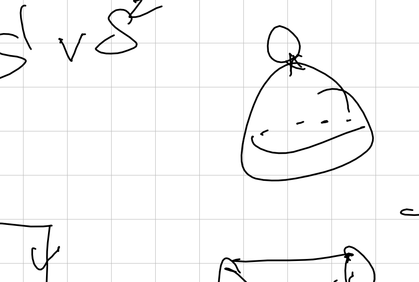
M: n-dim
with $(n\geq 3)$

then $\pi_1(M)\simeq\pi_1(M\setminus \bullet)$.
$\simeq\pi_1(M\setminus 2\text{pt.})$
$\simeq\pi_1(M\setminus\{n\})$.

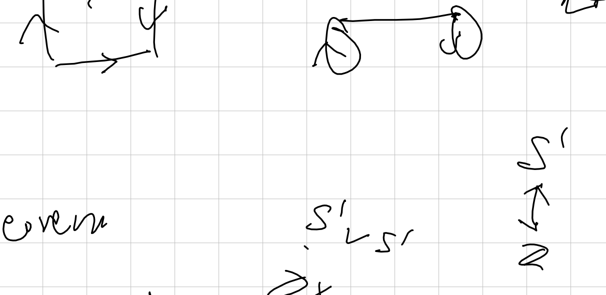
$\pi_1(S^n)=0$
$n\geq 2$.

$\pi_1(\mathbb{RP}^n)=\mathbb{Z}_2$
$(n\geq 2)$.

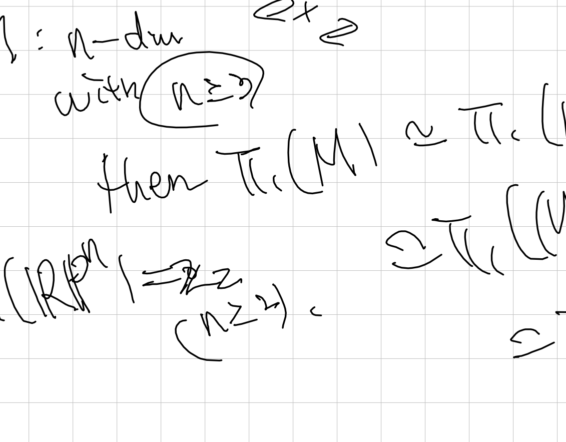
~~tree~~



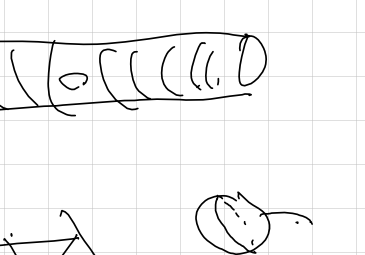
$S^1\vee S^1\vee\mathbb{R}$.
$\downarrow$
$\mathbb{Z}*\mathbb{Z}$



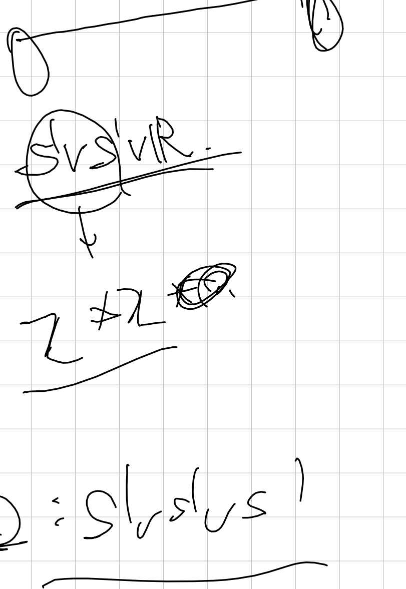
: $S^1\vee S^1\vee S^1$
$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$



Example.

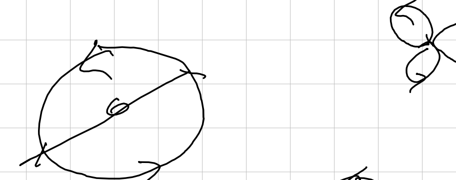


$S^2/N=S$ N



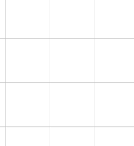
$\rightarrow\pi_1(\ \dots\ )$

$S^1\vee S^1$



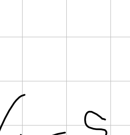
S^1: 0-cell 1개
1-cell 1개.

S^2: 0-cell 1개
2-cell 1개



$S^1\vee S^1$

$\mathbb{RP}^2$: 0-cell: 1개
1-cell: 1개
2-cell 1개

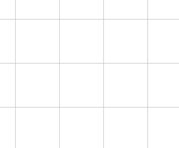


$aba^{-1}b^{-1}=0$.

$\langle a,b \mid ab=ba^{-1}\rangle$

$(\mathbb{Z}*\mathbb{Z})/a\dots$

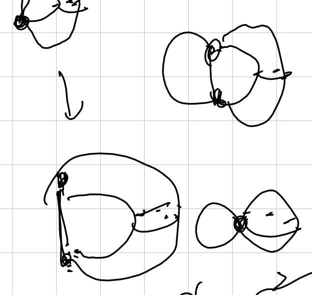
$\mathbb{RP}^2$: $\mathbb{Z}_2$ $\langle a \mid a^2\rangle$
$a^2=1$

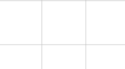


~~$\mathbb{RP}^3$: 0-cell 1개 1-cell 1개 2-cell 1개 3-cell 1개~~

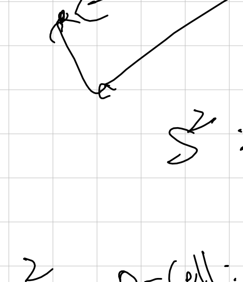
~~$\pi_1(\mathbb{RP}^\infty)$~~

$\pi_1(\mathbb{RP}^3)=\pi_1(\mathbb{RP}^2)=\mathbb{Z}_2$.

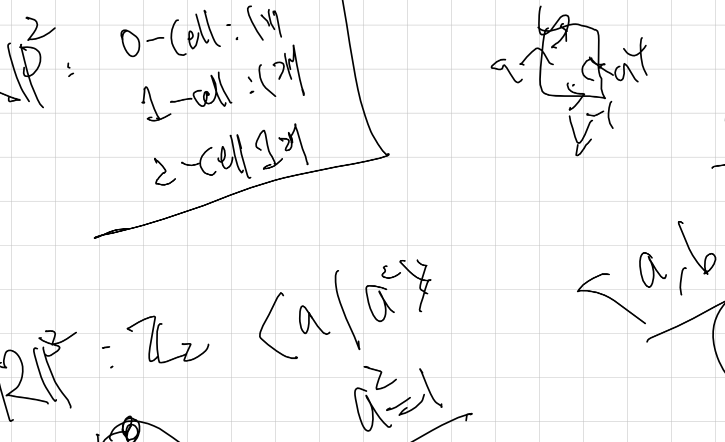




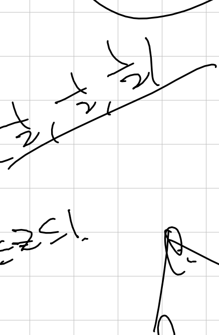
$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$

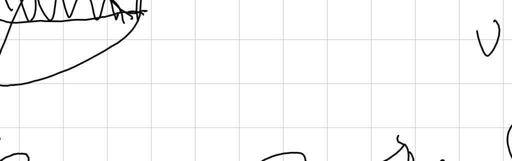


$\leftarrow$ $x^2+y^2+z^2=1$
$\cup$ $x^2+y^2\leq 1$
$\cup$ $(0,0,z)$ $-1\leq z\leq 1$.



$\mathbb{R}/\mathbb{Z}$





$z=x^4+y^4$
$\downarrow$ exp

$\frac{1}{p_n+z}$.

$K_1=x^4+z^2$
$K_2=x^4$ ..

$\int\frac{\dots dz}{y}$ $\oint\frac{1}{z}dz$
$2\pi i$

$2\pi i\cdot X(\dots)$
$\downarrow$
0